

Emmanuel Mwendwa Mayoli

AI Systems Builder & Electrical Engineering Student

 Nairobi, Kenya  lemanuelmwesh@gmail.com  0756061190  github.com/mega7306626007

 mega7306626007.github.io/portfolio

PROFILE

Engineering student and software builder with a strong focus on AI engineering, systems, mobile development, Python, and intelligent applications. Specialized in building practical, offline-first systems spanning AI, Android, language learning, finance, speech recognition, and creative technology. Experienced in developing 3 mobile apps and 4 websites alongside a growing portfolio of independent software projects.

TECHNICAL SKILLS

AI & SYSTEMS ENGINEERING

AI engineering, machine learning concepts, offline/local-first AI, conversational systems, ONNX inference, neural networks.

PROGRAMMING & FRAMEWORKS

Python, JavaScript, Kotlin / Android, Apps Script, Flask, Room Database, Speech/TTS, Git / GitHub.

LANGUAGES

English (Fluent), Kiswahili (Native), French (Conversational / Learning).

CREATIVE & WRITING

Published Author, Technical & Creative Writing, Poetry, Essays, Singing, Piano Performance.

EDUCATION

University of Nairobi (UoN) — BSc. Electrical & Electronic Engineering (EEE)

Current Student

Alliance High School — KCSE Secondary Education

KCSE: A Plain

Buruburu 1 Primary School — Primary Education

KCPE: 413 / 500

PUBLISHED CREATIVE WORKS



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Heart
Cover
Page

Author of "THE HEART" — An Anthology of Poems

Created and authored a full poetry collection exploring human emotion, reflection, and creative expression. Intersected literary work with tech projects by building *The Heart* v2 & v4—web platforms utilizing in-browser neural AI continuation for interactive reading experiences.

SELECTED SOFTWARE PROJECTS

Jarvis (Mwesh)

Kotlin / Android / ONNX

On-device multilingual voice assistant featuring ONNX inference, Room memory, command routing, speech recognition, TTS, and real-time translation across English, Kiswahili, and French.

PesaFlow

Kotlin / Android

Student-finance and adaptive-intelligence system designed around irregular income, campus economic pressures, financial planning, and offline-first data persistence.

Parlons

Kotlin / Android

Offline French-learning application tailored for Kenyan learners with interactive lessons, spaced repetition algorithms,

SELECTED SOFTWARE PROJECTS (CONTINUED)

PyChat

[Python / ML Stack](#)

Offline conversational AI engine combining persistent memory, intent handling, neural components, and an optional LLM-hybrid architecture; includes a publicly deployed demo.

Transcriber

[Python / Whisper AI](#)

Local Whisper-based audio transcription tool featuring synchronized audio playback, live SRT editing, Kiswahili language optimization, and a dedicated model fine-tuning workflow.

The Heart v4 & v2

[JavaScript / Python](#)

Interactive poetry web experiences featuring in-browser neural text continuation, combining creative literary writing with experimental AI and Python backend bridging.

Global Digest & Calendar Rescheduler

[Python / Flask / Apps Script](#)

Automated news engine structuring stories into newspaper digests; alongside a Google Calendar automation tool with push-to-free-slot scheduling, overfill management, and undo support.

EXPERIENCE & VOLUNTEERING

Sunday School Teacher

[KAG Life Springs, Kikuyu](#)

Assisted in teaching and mentoring youth students; developed strong interpersonal leadership, lesson delivery, and communication skills.

Volunteer Assistant

[Kenya National Library, Buruburu](#)

Shelved and systematically organized literary collections, assisted visitors, and maintained facility order and cleanliness.

Academic & Administrative Volunteer

[Lukenya Girls' High School](#)

Assisted staff with administrative duties, clerical tasks, campus maintenance support, and marking examination papers.

LEADERSHIP & ACHIEVEMENTS

- **Student Leadership (Alliance High School):** Organized school activities, led peer groups, and actively mentored junior students (2022–2025).
- **Certificate of Merit – IAAC (2025):** Awarded for outstanding performance in the International Astronomy and Astrophysics Competition.
- **Certificate of Merit – Robotics Competition:** Recognized at the regional robotics contest hosted at Lenana High School (2024).
- **Certificates of Merit – Société Polyvalente & Music Festivals:** Awarded for French Role Play excellence and 4 consecutive years of achievements in National Music Festivals.
- **Technical Certifications:** Certificate of Basics in Python; Certificate in Basics of Artificial Intelligence and Misinformation (Atingi).

INTERESTS & HOBBIES

 Piano Performance

 Poetry & Essay Writing

 Vocal Performance & Singing

 Stargazing & Astronomy

 Building AI Systems

 Mobile Software Engineering

References available upon request.